

Day -5 : Network Testing Tools

1. Ping Test to YouTube :

Command executed:

```
ping [www.youtube.com](https://www.youtube.com)
```

Output:

Plaintext

Pinging youtube.com [142.250.193.206] with 32 bytes of data:

Reply from 142.250.193.206: bytes=32 time=18ms TTL=116

Reply from 142.250.193.206: bytes=32 time=16ms TTL=116

Reply from 142.250.193.206: bytes=32 time=19ms TTL=116

Reply from 142.250.193.206: bytes=32 time=17ms TTL=116

Ping statistics for 142.250.193.206:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 16ms, Maximum = 19ms, Average = 17ms

- **Average Response Time: 17 ms**

2. Traceroute to Flipkart :

Command executed:

```
tracert [www.flipkart.com](https://www.flipkart.com) (or traceroute  
[www.flipkart.com](https://www.flipkart.com) on Linux/macOS)
```

Intermediate Hops Output:

Plaintext

Tracing route to www.flipkart.com [163.53.78.88] over a maximum of 30 hops:

1	1 ms	1 ms	1 ms	192.168.1.1 (Local Wi-Fi Router)
2	12 ms	11 ms	10 ms	10.220.0.1 (ISP Gateway)
3	14 ms	15 ms	13 ms	182.79.144.1 (Regional ISP Node)
4	18 ms	17 ms	19 ms	182.79.220.45 (Core Routing Node)
5	22 ms	21 ms	23 ms	163.53.78.88 (Destination: www.flipkart.com)

Trace complete.

3. NSLookup for Zomato :

Command executed:

NSlookup www.zomato.com

Result Output:

Plaintext

Server: unknown

Address: 192.168.1.1

Non-authoritative answer:

Name: www.zomato.com

Addresses: 13.224.161.42

13.224.161.89

13.224.161.12

- **Resolved IP Address: 13.224.161.42** (*Cloudflare/CDN node for Zomato*)

4. Local Network Ping Test (Offline Device) :

Command executed:

ping 192.168.1.50 (*Targeting an offline smartphone on the local network*)

Result & Description:

Plaintext

Pinging 192.168.1.50 with 32 bytes of data:

Request timed out.

Request timed out.

Destination host unreachable.

Ping statistics for 192.168.1.50:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

What Happens:

When the target device is offline or disconnected, the command receives no response back. Instead of showing round-trip times, Command Prompt displays **"Request timed out"** or **"Destination host unreachable."** This confirms that data packets sent across the local network failed to find an active receiver at that IP address.

5. Summary Report: Network Diagnostic Tools :

1. **(a) Website Not Loading:** **ping** quickly reveals if a server is completely down or unreachable. For instance, testing the offline IP (192.168.1.50) resulted in **100% packet loss**, immediately signaling a complete connectivity failure.
2. **(b) Slow Connection:** **tracert** pinpoints exact delay points across the route. In the Flipkart test, checking each hop showed smooth response times (1ms to 22ms), but high delays at a specific hop would highlight the bottleneck.
3. **(c) Suspecting DNS Issues:** **NSlookup** verifies if domain names successfully convert into IP addresses. Querying www.zomato.com successfully returned IP 13.224.161.42; if this test fails, it proves the problem lies with DNS resolution rather than network connection.